Federated Learning: A Gentle Introduction

Federated learning is a machine learning technique that trains an algorithm across multiple decentralized edge devices or servers holding local data samples, without exchanging the raw data itself.

This approach stands in contrast to traditional centralized machine learning techniques where all data samples are uploaded to one server, as well as to more classical decentralized approaches which often assume that local data samples are identically distributed.

How a training round works

First, the central server sends the current global model to a subset of participating devices. Each device then trains the model locally using its own private data, without that data ever leaving the device.

Next, each device computes a model update, such as a gradient or a set of weights, and sends only that update back to the central server, not the underlying data.

Finally, the central server aggregates the updates from all participating devices, typically by averaging them, to produce an improved global model. This cycle repeats for many rounds until the model converges.